Aleksandr Lobanov

Curriculum Vitae

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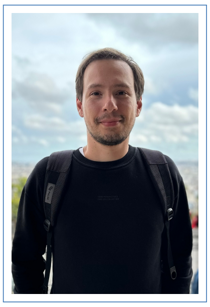

About Me

I am a Research Scientist at the Artificial Intelligence Center of Lomonosov Moscow State University. My research focuses on the mathematical foundations of optimization for machine learning, including stochastic optimization, generalized smoothness, training dynamics, derivative-free optimization, and distributed learning. A central theme of my recent work is understanding modern optimization mechanisms — such as gradient clipping, normalization, and learning-rate warmup — beyond classical global smoothness assumptions. I received my PhD in Applied Mathematics and Computer Science from the Moscow Institute of Physics and Technology in 2025 under the supervision of Prof. Alexander Gasnikov. In 2024, I was also a Visiting Researcher in the Department of Machine Learning at Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), hosted by Prof. Martin Takáč.

Education

2022–2025 : **PhD, Applied Mathematics & Computer Science**, *Moscow Institute of Physics and Technology*, Dolgoprudny, Russia.
I defended my PhD thesis, "Gradient-Free Methods for Convex Optimization under Noise Conditions," under the supervision of Alexander Gasnikov.

2020–2022 : **Master of Applied Mathematics & Computer Science**, *Moscow Aviation Institute*, Moscow, Russia.
I did my master's thesis "Development and Application of Mini-batch Adaptive Optimization Algorithms in Joint Estimation and Control Problems" supervised by Andrei Panteleev.

2016–2020 : **Bachelor of Applied Mathematics**, *Moscow Aviation Institute*, Moscow, Russia.
I did my bachelor's thesis "Optimization methods in machine learning for the identification of dynamic systems parameters" supervised by Andrei Panteleev.

Work Experience

Jan., 2026 – present: ***Lomonosov Moscow State University***, *Artificial Intelligence Center*.
Position: *Research Scientist*; Group Head: *Yuriy Dorn*

Sep., 2024 – present: ***Higher School of Economics***, *Laboratory for Theoretical Foundations of AI Models*.
Position: *Junior Researcher*; Group Head: *Vladimir Spokoiny*

Aug., 2022 – Apr., 2026: ***Moscow Institute of Physics and Technology***, *Laboratory of Advanced Combinatorics and Network Applications*.
Position: *Researcher*; Group Head: *Andrei Raigorodskii*

Apr., 2023 – Dec., 2025: ***Ivannikov Institute for System Programming of the RAS***, *Research Center for Trusted Artificial Intelligence*.
Position: *Research Intern*; Group Head: *Denis Turdakov*

May, 2024 – May, 2025 ***Skolkovo Institute of Science and Technology***, *Laboratory "Multiscale Neurodynamics for Intelligent Systems"*.
Position: *Junior Researcher*; Group Head: *Evgeny Burnaev*

Nov., 2024 – Dec., 2024 ***Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)***, *Department of Machine Learning*.
Position: *Visiting Researcher*; Group Head: *Martin Takáč*

Publications

2026 **Aleksandr Lobanov** and Anastasia Koloskova. Avoiding bias in clipped sgd for overparameterized models under generalized smoothness. *arXiv preprint arXiv:2605.14800*, 2026.

2026 **Aleksandr Lobanov**. A few accelerated algorithms for convex optimization under (h_0, h_1)-smoothness. *arXiv preprint arXiv:2608.04884*, 2026.

2026 Nail Bashirov, Alexander Gasnikov, and **Aleksandr Lobanov**. Zeroth-order methods for non-smooth stochastic problems under heavy-tailed noise. *Optimization Methods and Software*, pages 1–26. Taylor & Francis, 2026.

2025 **Aleksandr Lobanov** and Alexander Gasnikov. Survey of modern smooth optimization algorithms with comparison oracle. In *Doklady Mathematics*, volume 526, pages 16–23, 2025.

2025 **Aleksandr Lobanov** and Alexander Gasnikov. Power of generalized smoothness in stochastic convex optimization: First-and zero-order algorithms. *arXiv preprint arXiv:2501.18198*, 2025.

2025 **Aleksandr Lobanov**. *Gradient-Free Methods for Convex Optimization under Noise Conditions*. PhD thesis, Moscow Institute of Physics and Technology, 2025.

2024 **Aleksandr Lobanov**, Alexander Gasnikov, and Fedor Stonyakin. Highly smooth zeroth-order methods for solving optimization problems under the pl condition. *Computational Mathematics and Mathematical Physics*, volume 64, pages 739–770. Springer, 2024.

2024 **Aleksandr Lobanov**, Alexander Gasnikov, and Andrei Krasnov. Acceleration exists! optimization problems when oracle can only compare objective function values. *Advances in Neural Information Processing Systems*, volume 37, pages 21428–21460, 2024.

2024 **Aleksandr Lobanov**, Alexander Gasnikov, Eduard Gorbunov, and Martin Takáč. Linear convergence rate in convex setup is possible! gradient descent method variants under (l_0, l_1)-smoothness. *arXiv preprint arXiv:2412.17050*, 2024.

2024 **Aleksandr Lobanov** and Alexander Gasnikov. Improved maximum noise level estimation in black-box optimization problems. *Journal of Mathematical Sciences*, volume 540, pages 132–147, 2024.

2024 **Aleksandr Lobanov**, Nail Bashirov, and Alexander Gasnikov. The “black-box” optimization problem: Zero-order accelerated stochastic method via kernel approximation. *Journal of Optimization Theory and Applications*, volume 203, pages 2451–2486. Springer, 2024.

2024 **Aleksandr Lobanov**. Improved iteration complexity in black-box optimization problems under higher order smoothness function condition. *arXiv preprint arXiv:2407.03507*, 2024.

2024 Ekaterina Statkevich, Sofiya Bondar, Darina Dvinskikh, Alexander Gasnikov, and **Aleksandr Lobanov**. Gradient-free algorithm for saddle point problems under overparametrization. *Chaos, Solitons & Fractals*, volume 185, page 115048. Elsevier, 2024.

2024 Vladimir Smirnov, Kristina Kazistova, Ilya Sudakov, Valentin Leplat, Alexander Gasnikov, and **Aleksandr Lobanov**. Asymptotic analysis of the ruppert polyak averaging for stochastic order oracle. *Russian Journal of Nonlinear Dynamics*, volume 20, pages 961–978, 2024.

2024 Nikita Kornilov, Yuriy Dorn, **Aleksandr Lobanov**, Nikolay Kutuzov, Innokentiy Shibaev, Eduard Gorbunov, Alexander Nazin, and Alexander Gasnikov. Median clipping for zeroth-order non-smooth convex optimization and multi-armed bandit problem with heavy-tailed symmetric noise. *arXiv preprint arXiv:2402.02461*, 2024.

2024 Alexander Gasnikov, Mohammad Alkousa, **Aleksandr Lobanov**, Yutii Dorn, Fedor Stonyakin, Ilya Kuruzov, and Sanjeev Singh. On quasi-convex smooth optimization problems by a comparison oracle. *Russian Journal of Nonlinear Dynamics*, volume 20, pages 813–825, 2024.

2024 Boris Chervonenkis, Andrei Krasnov, Alexander Gasnikov, and **Aleksandr Lobanov**. Nesterov's method of dichotomy via order oracle: The problem of optimizing a two-variable function on a square. In *International Conference on Optimization and Applications*, pages 58–68. Springer, 2024.

2024 Georgii Bychkov, Darina Dvinskikh, Anastasiya Antsiferova, Alexander Gasnikov, and **Aleksandr Lobanov**. Accelerated zero-order sgd under high-order smoothness and overparameterized regime. *Russian Journal of Nonlinear Dynamics*, volume 20, pages 759–788, 2024.

2024 Seydamet Ablaev, Aleksandr Beznosikov, Alexander Gasnikov, Darina Dvinskikh, **Aleksandr Lobanov**, Sergei Puchinin, and Fedor Stonyakin. On some works of boris teodorovich polyak on the convergence of gradient methods and their development. *Computational Mathematics and Mathematical Physics*, volume 64, pages 635–675. Springer, 2024.

2023 Daniil Vostrikov, Georgiy Konin, **Aleksandr Lobanov**, and Vladislav Matyukhin. Influence of the mantissa finiteness on the accuracy of gradient-free optimization methods. *Computer research and modeling*, volume 15, pages 259–280, 2023.

2023 **Aleksandr Lobanov**, Andrew Veprikov, Georgiy Konin, Aleksandr Beznosikov, Alexander Gasnikov, and Dmitry Kovalev. Non-smooth setting of stochastic decentralized convex optimization problem over time-varying graphs: A. lobanov et al. *Computational Management Science*, volume 20, page 48. Springer, 2023.

2023 **Aleksandr Lobanov** and Alexander Gasnikov. Accelerated zero-order sgd method for solving the black box optimization problem under "overparametrization" condition. In *International Conference on Optimization and Applications*, pages 72–83. Springer, 2023.

2023 **Aleksandr Lobanov**, Anton Anikin, Alexander Gasnikov, Alexander Gornov, and Sergey Chukanov. Zero-order stochastic conditional gradient sliding method for non-smooth convex optimization. In *International Conference on Mathematical Optimization Theory and Operations Research*, pages 92–106. Springer, 2023.

2023 **Aleksandr Lobanov**, Belal Alashqar, Darina Dvinskikh, and Alexander Gasnikov. Gradient-free federated learning methods with l_1 and l_2-randomization for non-smooth convex stochastic optimization problems. *Computational Mathematics and Mathematical Physics*, volume 63, pages 1600–1653. Springer, 2023.

2023 **Aleksandr Lobanov**. Stochastic adversarial noise in the "black box" optimization problem. In *International Conference on Optimization and Applications*, pages 60–71. Springer, 2023.

2023 Sergey Sadykov, **Aleksandr Lobanov**, and Andrei Raigorodskii. Gradient-free algorithms for solving stochastic saddle optimization problems with the polyak–łojasiewicz condition. *Programming and Computer Software*, volume 49, pages 535–547. Springer, 2023.

2023 Dmitry Pasechnyuk, **Aleksandr Lobanov**, and Alexander Gasnikov. Upper bounds on maximum admissible noise in zeroth-order optimisation. *arXiv preprint arXiv:2306.16371*, 2023.

2023 Nikita Kornilov, Ohad Shamir, **Aleksandr Lobanov**, Darina Dvinskikh, Alexander Gasnikov, Innokentiy Shibaev, Eduard Gorbunov, and Samuel Horváth. Accelerated zeroth-order method for non-smooth stochastic convex optimization problem with infinite variance. *Advances in Neural Information Processing Systems*, volume 36, pages 64083–64102, 2023.

2023 Alexander Gasnikov, Darina Dvinskikh, Pavel Dvurechensky, Eduard Gorbunov, Aleksandr Beznosikov, and **Aleksandr Lobanov**. Randomized gradient-free methods in convex optimization. In *Encyclopedia of Optimization*, pages 1–15. Springer, 2023.

2023 Jiawei Chen, **Aleksandr Lobanov**, and Alexander Rogozin. Distributed min-max optimization using the smoothing technique. *Computer Research and Modeling*, volume 15, pages 469–480, 2023.

2022 Andrei Panteleev and **Aleksandr Lobanov**. Application of the zero-order mini-batch optimization method in the tracking control problem. In *SMART Automatics and Energy: Proceedings of SMART-ICAE 2021*, pages 573–581. Springer, 2022.

2022 Andrei Panteleev and **Aleksandr Lobanov**. Application of mini-batch adaptive optimization method in stochastic control problems. In *Advances in Theory and Practice of Computational Mechanics: Proceedings of the 22nd International Conference on Computational Mechanics and Modern Applied Software Systems (CMMASS 2021)*, pages 345–361. Springer, 2022.

2021 Andrei Panteleev and **Aleksandr Lobanov**. Application of the mini-batch adaptive method of random search (mamrs) in problems of optimal in mean control of the trajectory pencils. In *Journal of Physics: Conference Series*, volume 1925, page 012006. IOP Publishing, 2021.

2021 Andrei Panteleev and **Aleksandr Lobanov**. Application of mini-batch metaheuristic algorithms in problems of optimization of deterministic systems with incomplete information about the state vector. *Algorithms*, volume 14, page 332. MDPI, 2021.

2020 Andrei Panteleev and **Aleksandr Lobanov**. Mini-batch adaptive random search method for the parametric identification of dynamic systems. *Automation and Remote Control*, volume 81, pages 2026–2045. Springer, 2020.

2020 Andrei Panteleev and **Aleksandr Lobanov**. The mini-batch adaptive method of random search (mamrs) for parameters optimization in the tracking control problem. In *IOP Conference Series: Materials Science and Engineering*, volume 927, page 012025. IOP Publishing, 2020.

2019 Andrei Panteleev and **Aleksandr Lobanov**. Gradient optimization methods in machine learning for the identification of dynamic systems parameters. *Modelling and Data Analysis*, page 98, 2019.

Talks & Scientific Events

September 28–29, 2026 ***"Understanding the Effectiveness of Gradient Clipping and Warmup: Two-Phase Convergence under Generalized Smoothness".*** Invited Talk, 2nd Wireless Workshop “Step into the Future”, Wireless Business Unit, Huawei Russian Research Institute. Irkutsk, Russia (scheduled).

July, 20-25, 2026 ***The 16th Lisbon Machine Learning School (LxMLS) 2026.*** Summer School at Instituto Superior Técnico (IST). Lisbon, Portugal.

May 30, 2026 ***"Gradient-Free Oracle as a Black Box: How to Solve Optimization Problems When the Oracle Has No Access to Derivatives".*** Invited Research Seminar, Zhi-Quan (Tom) Luo's Group, The Chinese University of Hong Kong, Shenzhen. Shenzhen, China (online).

March. 27 - April 02, 2026 ***SLAI Academic Exchange.*** Academic Exchange Activities at Shenzhen Loop Area Institute. Shenzhen, China.

October 07, 2025 ***"On the Possibility of Linear Convergence in the Convex Setting: GD-type First- and Zero-Order Optimization Algorithms under Generalized Smoothness".*** Quasilinear Equations, Inverse Problems and their Applications (QIPA 2025). Sirius, Russia.

September 30, 2025 ***"Power of Generalized Smoothness in Convex Optimization: First- and Zero-Order Algorithms under Generalized Smoothness".*** Scientific and Practical Intensive on Replication of State-of-the-art Scientific Results. Sirius, Russia.

Sep. 08 - 12, 2025 ***Mediterranean Machine Learning Summer School.*** Summer School at University of Split. Split, Croatia.

August 01, 2025 ***"Linear Convergence Rate in Convex Setup is Possible! First- and Zero-Order Algorithms under Generalized Smoothness".*** XVII th Conference on Stochastic Programming (ICSP 2025). Champs-sur-Marne, France.

June 23 - 28, 2025 ***Summer School "Control, Information and Optimization".*** Summer School at Innopolis University. Innopolis, Russia.

May 25, 2025 ***"Linear Convergence Rate in Convex Setup is Possible! Gradient Descent Method Variants under (L0,L1)-Smoothness".*** Data Fest 2025 (Yandex). Moscow, Russia.

March 28, 2025 ***"Linear Convergence Rate in Convex Setup is Possible! First- and Zero-Order Algorithms under Generalized Smoothness".*** The International Conference dedicated to mathematics in artificial intelligence. Sirius, Russia.

Nov.01-Dec.28, 2024 ***Internship Program at Mohamed bin Zayed University of Artificial Intelligence (MBZUAI).*** Research Assistant at MBZUAI. Abu Dhabi, UAE.

October 26, 2024 ***"Acceleration Exists! Optimization Problems When Oracle Can Only Compare Objective Function Values".*** Conference on Machine Learning "Fall into ML 2024". Moscow, Russia.

October 19, 2024 ***"Advantages of the batching technique in gradient-free optimization problems".*** Quasilinear Equations, Inverse Problems and their Applications. Sirius, Russia.

October 16, 2024 ***"A survey of techniques for creating randomized gradient-free algorithms for solving convex optimization problems".*** Scientific and Practical Intensive on Replication of State-of-the-art Scientific Results. Sirius, Russia.

October 14, 2024 ***"Acceleration Exists! Optimization Problems When Oracle Can Only Compare Objective Function Values".*** Inverse incorrect problems and machine learning. Sirius, Russia.

October 11, 2024 ***"Convergence of gradient-free algorithms under overparameterization conditions for federated learning architecture".*** Trusted artificial intelligence. Sirius, Russia.

October 07-09, 2024 ***AI Autumn School on Computational Optimization.*** Autumn School at Innopolis University. Innopolis, Russia.

September 27, 2024 ***"Gradient-free oracle as a black box: how to solve optimization problems when the oracle has no access to derivatives".*** Optimization & Decision Intelligence Group Internal Seminar. Zurich, Switzerland (online).

September 25, 2024 ***"State-of-the-art Zero-Order Randomized Algorithms for Convex Optimization".*** Research and Practice Workshop: At the center of AI. Saint Petersburg, Russia.

August 29-30, 2024 ***"Acceleration Exists! Optimization Problems When Oracle Can Only Compare Objective Function Values".*** ALGOPT2024 workshop on Algorithmic Optimization: Tools for AI and Data Science. Louvain-la-Neuve, Belgium.

August 11, 2024 ***"AI Wine, Chocolate, etc. Or How to Solve Optimization Problems When Only the Objective Function Values Can be Compared?".*** IV Conference of Russian Mathematical Centers. Saint Petersburg, Russia.

Jul. 29 - Aug. 3, 2024 ***Summer School "Control, Information and Optimization".*** Summer School at Novosibirsk State University. Novosibirsk, Russia.

July, 11-17, 2024 ***The 14th Lisbon Machine Learning School (LxMLS) 2024.*** Summer School at Instituto Superior Técnico (IST). Lisbon, Portugal.

June 27, 2024 ***"State-of-the-art Gradient-Free Randomized Convex Optimization Algorithms".*** Main scientific seminar of the Innopolis University «Innopolis. Science». Innopolis, Russia.

June 25, 2024 ***"The "Overbatching" Effect? Yes, or How to Improve Error Floor in Black-Box Optimization Problems".*** Device Algorithm Summit 2024 (Huawei). Moscow, Russia.

June 19, 2024 ***"Presentation of Papers Submitted to the NeurIPS Conference 2024".*** Institute of Artificial Intelligence (MSU) internal seminar. Moscow, Russia.

April 24, 2024 ***"Advancing Maximum Noise Level Estimation in the Black-Box Optimization Problems".*** WAIT: Workshop on Artificial Intelligence Trustworthiness. Almaty, Kazakhstan.

February 08, 2024 ***"How to Solve Optimization Problems When You Can Only Compare Objective Function Values? Applications for Creating AI Wines, Chocolates, etc.".*** Spring into ML 2024. Innopolis, Russia.

December 04, 2023 ***"Accelerated Zeroth-order Method for Non-Smooth Stochastic Convex Optimization Problem with Infinite Variance".*** Ivannikov ISP RAS Open Conference. Moscow, Presidium of RAS, Russia.

November 07, 2023 ***"Techniques for creating gradient-free randomized algorithms in different settings of the optimization problem".*** Trusted artificial intelligence. Sirius, Russia.

October 13, 2023 ***"A survey of techniques for creating randomized gradient-free algorithms for solving convex optimization problems (under the overparameterization condition)".*** Scientific and Practical Intensive on Replication of State-of-the-art Scientific Results. Sirius, Russia.

September 20, 2023 ***"Accelerated Zero-Order SGD Method for Solving the Black Box Optimization Problem under "Overparametrization" Condition".*** XIV International Conference Optimization and Applications (OPTIMA-2023). Petrovac, Montenegro.

September 20, 2023 ***"Stochastic Adversarial Noise in the "Black Box" Optimization Problem".*** XIV International Conference Optimization and Applications (OPTIMA-2023). Petrovac, Montenegro.

September 15, 2023 ***"Survey of modern randomized gradient-free algorithms for convex optimization problems".*** All-Russian seminar on optimization named after B.T. Polyak. Dolgoprudny, Russia (online).

July, 10-15, 2023 ***Summer School Learning, Understanding and Optimization in Artificial Intelligence Models.*** Summer School Control, Information and Optimization. Nizhny Novgorod, Russia.

July 04, 2023 ***" Zero-Order Algorithms for Solving Non-Smooth Stochastic Optimization Problems with Heavy-Tailed Noise".*** Mathematical Optimization Theory and Operations Research. Ekaterinburg, Russia.

July 04, 2023 ***"Zero-Order Stochastic Conditional Gradient Sliding Method for Non-smooth Convex Optimization".*** Mathematical Optimization Theory and Operations Research. Ekaterinburg, Russia.

May 11, 2023 ***"Non-Smooth Setting of the Stochastic Decentralized Convex Optimization Problem Over Time-Varying Graphs".*** The 13th International Conference on Network Analysis. Nizhny Novgorod, Russia (online).

April 22, 2023 ***"Stochastic Adversarial Noise in the «Black Box» Optimization Problem".*** WAIT: Workshop on Artificial Intelligence Trustworthiness. Yerevan, Armenia.

April 5, 2023 ***"Is a smoothing scheme with l_1 randomization better?".*** 65th Russian Scientific Conference of MIPT. Dolgoprudny, Russia.

August 22, 2022 ***"Two-Point and One-Point Gradient-Free Methods for Federated Learning".*** Quasilinear Equations, Inverse Problems and Their Applications. Sochi, Russia.

June, 20-26, 2022 ***Summer School Learning, Understanding and Optimization in Artificial Intelligence Models.*** Summer School at HSE University. Pushkin, Russia.

July 22, 2021 ***"Solving the command navigation problem using a mini-batch adaptive optimization method applied in machine learning".*** The International Conference "Mathematical Modelling". Zhukovsky, Russia (online).

Jul. 19 - Aug. 8, 2021 ***Modern Methods of Information Theory, Optimization and Control.*** Intern-Researcher at Sirius University of Science and Technology. Sochi, Russia.

April 26-30, 2021 ***Big Challenges for Society, State, and Science.*** Summit of Young Scientists and Engineers. Sochi, Russia.

April 17, 2020 ***"Development and application of optimization methods in machine learning procedures for finding parameters of continuous dynamic systems".*** Gagarin Science Conference. Moscow, Russia.

Teaching Assistantship

Fall 2023 : **Optimization Methods**.
This one-semester course is for the third year bachelor's students at DIHT, MIPT.

Spring 2023 : **Optimization Methods**.
This one-semester course is for the second year bachelor's students at MAI.

Fall 2022 : **Optimization Methods**.
This one-semester course is for the third year bachelor's students at DIHT, MIPT.

Fall 2022 : **Discrete Mathematics**.
This one-semester course is for the second year bachelor's students at MAI.

Fellowships & Awards

2024 – 2025 ***Scholarship of the President of the Russian Federation for PhD students and adjuncts*** (given for the achievements in research activities), Russia

2023 – 2025 ***B.T. Polyak Scholarship for Contribution to the Development of Numerical Methods of Optimization*** (given for the achievements in research activities), Russia

2024 ***NeurIPS travel award*** (given for the achievements in research activities), Vancouver Convention Center, Canada

2020 – 2022 ***Grant of the President of the Russian Federation for master's degree students*** (given for the achievements in research activities), Russia

2020 – 2022 ***Scholarship of the Government of the Russian Federation*** (given for the achievements in research activities), Russia

2020 ***First Place Diploma*** (awarded for speaking at the Gagarin Science Conference), Russia

Reviewing Activities

Area Chair The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026).
Reviewer Annual AAAI Conference on Artificial Intelligence (AAAI'27).
Reviewer International Conference on Machine Learning (ICML'25,26).
Reviewer International Conference on Learning Representations (ICLR'25,26).
Reviewer Special Interest Group on Knowledge Discovery and Data Mining (SIGKDD'26).
Reviewer Conference on Neural Information Processing Systems (NeurIPS'24,25).
Reviewer International Conference on Optimization and Applications (OPTIMA'23,24).
Reviewer International conference "Mathematical Optimization Theory and Operations Research" (MOTOR'23,24).
Reviewer Work- shop on Artificial Intelligence Trustworthiness (WAIT'23,24).

Computer skills

Programming: Python, PyTorch, C, C++, C#, MATLAB (basic), SQL

Hobbies

Football, Basketball, Cycling, Hiking, Swimming, Skiing, Solving Puzzle Games, Travelling